

Today 11:39 PM

**Week 2 research method...**
Document

Just like week 1, do for this.

Worked for 23s >

Absolutely. I went through the **full Week 2 transcript**, including Lectures 7–12, rather than relying only on the truncated preview. I'll keep the same structure as Week 1, but this time I'll also be a little more careful where the lecturer's logical/philosophical claims need qualification.

Week 2 — Logical Reasoning: Inductive, Deductive & Syllogistic Logic

Source: *Week 2 Research Methodology Transcript*, Module 2, Lectures 7–12.

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Lectures covered:

1. Lecture 7 — Logical Reasoning: **Inductive Logic**
2. Lecture 8 — Logical Reasoning: **Deductive Logic — Part 1**
3. Lecture 9 — Logical Reasoning: **Deductive Logic — Part 2**
4. Lecture 10 — Logical Reasoning: **Syllogistic Logic — Part 1**
5. Lecture 11 — Logical Reasoning: **Syllogistic Logic — Part 2**
6. Lecture 12 — Logical Reasoning: **Syllogism Logic, Truth and Validity**

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0. Source / Scope Header

Week

2

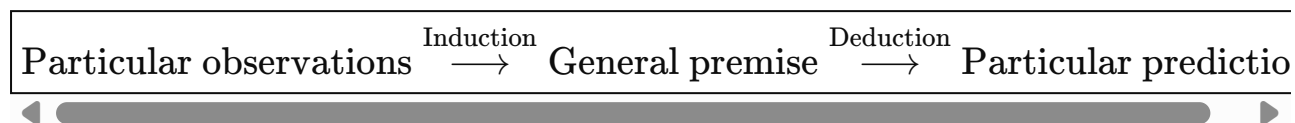
Module

Module 2 — Research Methodology

Central theme

How scientists move from observations to general principles, and then use those principles to reason about particular cases.

The week develops a logical cycle:



Then:

Prediction → Test → New observation → Revision
--

This is the core intellectual architecture of Week 2.

1. Lecture Map

Lecture	Central question	Main concepts
7	How do we move from observations to general knowledge?	Induction, inductive leap, uniformity of nature, causality, deterministic/probabilistic induction, enumeration, proper/improper induction, quantitative induction
8	How do we apply a general premise to a particular case?	Deduction, if-then logic, modus ponens, modus tollens, chains of deduction
9	What happens when conclusions are probabilistic or premises conflict?	Probabilistic deduction, probabilistic modus ponens, limits of probabilistic modus tollens, defeasible reasoning, scientific ethics
10	How can categorical arguments be structurally organized?	Syllogism, major/minor premise, middle term, A/E/I/O propositions
11	How can we determine whether syllogistic conclusions are valid?	Categorical syllogism, Venn diagrams, valid/invalid inference, possibility vs certainty

Lecture	Central question	Main concepts
12	Are truth and logical validity the same thing?	Truth vs validity, true/false premises, valid/invalid arguments, hypothesis testing, induction + deduction, Cepheid variables

2. Executive Understanding

What You Should Understand After Week 2

1. Induction goes from particular → general

You observe individual cases and infer a general statement.

Particular → General

Example:

Observe many insects with six legs → infer that insects have six legs.

The problem is that you have **not observed every possible insect**.

Therefore an **inductive leap** exists.  Week 2 research methodology tra...

2. Deduction goes from general → particular

You already have a general premise and apply it to a specific case.

General → Particular

Example:

All mammals have hot blood.

A leopard is a mammal.

Therefore, a leopard has hot blood.

The lecturer emphasizes that deductive reasoning requires a general premise already available.  Week 2 research methodology tra...

3. Induction is uncertain

Because induction goes beyond what has actually been observed, there is always an element of uncertainty.

The lecturer repeatedly emphasizes this point.  Week 2 research methodology tra...

4. Deduction can be logically certain—but only relative to its premises

If:

$$A \rightarrow B$$

and:

$$A$$

then:

$$B$$

This is **modus ponens**.

But this logical certainty does **not** mean the original premises are true.

That distinction becomes the centerpiece of Lecture 12.  Week 2 research methodology tra...

5. Truth ≠ validity

This is probably the **single most important conceptual distinction of Week 2**.

An argument can have:

- true premises + valid logic → true conclusion
- false premises + valid logic → potentially true conclusion
- true premises + invalid logic → false conclusion

Therefore:

Truth of statements ≠ Validity of reasoning

The lecturer demonstrates this explicitly using examples involving dinosaurs, mammals, electrons and insects.  Week 2 research methodology tra...

6. Science uses both induction and deduction

Science does not operate purely through one logical direction.

It repeatedly cycles between:

Induction → General model → Deduction → Prediction → Observation

The lecturer explicitly concludes that science progresses through the joint use of inductive and deductive logic.  Week 2 research methodology tra...

3. The Big Picture

Week 1 introduced:

What science is, why objectivity matters, and why causality matters.

Week 2 now asks:

How does the actual logical machinery of scientific reasoning work?

The progression is:



```
WEEK 1
Reality
↓
Observation
↓
Question
↓
Causality
↓
Scientific investigation

WEEK 2
Observation
↓
INDUCTION
↓
General premise / hypothesis
↓
DEDUCTION
↓
Prediction
↓
Testing
↓
New evidence
```

↓
Revision

So Week 2 is essentially the **logic engine** underneath scientific research.

4. Lecture 7 — Inductive Logic

4.1 What Is Inductive Reasoning?

The lecturer defines induction as movement:

Particular → General

You observe individual instances and generalize.

For example:

Observation

Clouds repeatedly precede rain.

Generalization

When clouds appear, rain may follow.

New case

Clouds appear today.

Prediction

It may rain today.

The first transition is **induction**.

The second is **deduction**.  Week 2 research methodology tra...

4.2 Induction Is at the Base of Science

The lecturer argues that scientific laws arise through generalization from observations.

Historically, Francis Bacon emphasized large-scale observation, data collection and generalization.  Week 2 research methodology tra...

The basic structure is:

$$O_1, O_2, O_3, \dots, O_n \rightarrow G$$

where:

- O_i = observations
 - G = generalization
-

4.3 Examples of Inductive Inference

The lecturer gives examples from different scientific fields:

Biology

All insects have six legs.

Chemistry

Copper turns green when dipped in vinegar.

Physics

Planets have elliptical orbits.

Geology

Volcanoes are generally associated with tectonic plate boundaries.

The point is not the individual claims themselves.

The point is:

How could scientists know a universal/general statement from a finite number of observations?

They cannot observe every possible instance.

Therefore:

Finite observations $\rightarrow$ general claim

contains an **inductive leap**.  Week 2 research methodology tra...

4.4 The Inductive Leap

Suppose you observe:

1000

insects.

Every one has six legs.

You infer:

All insects have six legs.

But logically:

1000 observations $\nRightarrow$ all possible insects

The conclusion contains information that was not literally contained in the observations.

That additional step is:

Inductive leap

The lecturer explicitly identifies this as moving from the known to the unknown.

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4.5 Why Do We Make the Inductive Leap?

The lecturer gives two foundations:

1. Uniformity of Nature

The assumption is:

Nature does not behave completely erratically; similar things under similar circumstances tend to behave similarly.

2. Causality

The assumption is:

Events have causes, and when the relevant causal conditions recur, corresponding effects should recur.

Together:

Uniformity of nature + Causality $\rightarrow$ Basis for induction
--

The lecturer explicitly identifies these as the two bases for the inductive leap.

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4.6 Critical Philosophical Point

There is an important issue here.

The lecturer presents uniformity and causality as the justification for induction.

But philosophically, this immediately raises the classic **problem of induction**:

How do we justify assuming that nature will continue behaving as it has behaved in the past?

If you say:

"Because it has always worked before."

you are already using induction.

So the justification risks becoming circular.

This is a deeper philosophical problem that Week 2 is implicitly introducing.

Important takeaway

Do not memorize:

"Uniformity of nature proves induction."

Instead understand:

Uniformity of nature and causal regularity are assumptions that make inductive scientific reasoning reasonable/useful, but the philosophical justification of induction is itself a difficult problem.

4.7 Deterministic vs Probabilistic Induction

The lecturer distinguishes two forms.

Deterministic

You say:

Unless a contrary example is found, the general statement can be accepted.

Example:

All insects have six legs, unless a genuine counterexample is found.

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Probabilistic

Instead of claiming certainty, you say:

There is a high degree of confidence that the statement is true.

Formally:

$$P(\text{claim is correct}) \approx 1$$

but:

$$P(\text{claim is wrong}) > 0$$

The lecturer emphasizes that a nonzero probability of error remains.

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4.8 The White Swan Example

Suppose you observe:

100

white swans.

You conclude:

All swans are white.

Then you discover a black swan.

Your generalization fails.

This demonstrates:

Many confirming observations $\nRightarrow$ universal certainty

The lecturer uses the swan example specifically to demonstrate this problem.

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4.9 Simple Enumeration

Suppose:

$$A_1 = B$$

$$A_2 = B$$

$$A_3 = B$$

and you conclude:

All A are B

This is **simple/incomplete enumeration**.

The lecturer calls the resulting inference doubtful because only a finite subset has been observed. 📄 Week 2 research methodology tra...

4.10 Complete Enumeration

Complete enumeration occurs when you genuinely inspect **all possible members** of a finite set.

Example:

Every month of the year has fewer than 32 days.

There are only 12 months.

You can inspect all 12.

Therefore:

Complete enumeration → No unexamined member remains

The lecturer contrasts this with scientific populations where complete enumeration is usually impossible. 📄 Week 2 research methodology tra...

4.11 Mathematical Induction

The lecturer then connects this to mathematical induction.

For a proposition $P(n)$:

Base case

$$P(1)$$

is true.

Inductive step

Show:

$$P(n) \Rightarrow P(n + 1)$$

Then:

$$P(1) \Rightarrow P(2) \Rightarrow P(3) \Rightarrow \dots$$

Therefore:

$P(n)$ is true for all positive integers
--

CRITICAL DISTINCTION

Mathematical induction is **not the same thing as empirical induction**.

Despite sharing the word "induction," mathematical induction is a **deductive proof method**: once the base case and implication are established, the conclusion follows logically.

The transcript describes it under complete enumeration, but mathematically it is better understood as deduction.

This is an important distinction to retain.

4.12 Improper vs Proper Induction

The lecturer distinguishes:

Improper induction

Generalization based simply on observed cases.

Example:

I've seen many white swans → all swans are white.

Proper induction

The generalization is supported by an understood causal connection.

Example:

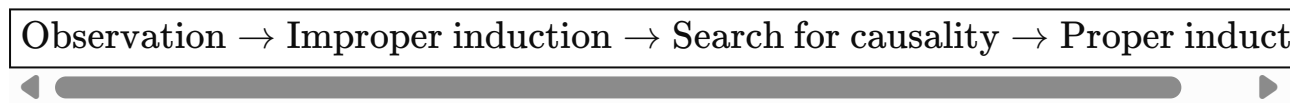
We observe copper changing under certain conditions and understand the mechanism explaining the change.

The lecturer therefore places **causal investigation** inside inductive reasoning.

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4.13 Research Progression

The lecturer describes the progression:



This connects directly with Week 1's discussion of causality.

4.14 Quantitative Induction

Sometimes the goal is not:

"All A are B."

Instead:

"What is the average/value/proportion of B?"

Example:

What is the average weight of adult male sparrows?

You cannot weigh every sparrow.

You take a sample:

$$X_1, X_2, \dots, X_n$$

and estimate a population quantity.

The lecturer calls this **quantitative inductive logic**.  Week 2 research methodology tra...

4.15 Population vs Sample

This distinction is fundamental.

Population

The entire group we want to know about.

Sample

The subset we actually observe.

Sample → Inference about population

This is the logical foundation of statistical inference.

4.16 Confidence / Probability

The lecturer gives an example where the population parameter is expressed as a range:

$$m_1 \leq X \leq m_2$$

with some probability/confidence.

The important methodological point is:

We do not pretend the sample gives us perfect knowledge of the entire population.

Instead, uncertainty is explicitly represented.

4.17 Proportion as an Inductive Quantity

Gregor Mendel's pea experiments are used as an example.

The lecturer discusses obtaining a fraction/proportion from observed generations.

Conceptually:

Sample observations → Estimated proportion

rather than merely a yes/no universal statement.  Week 2 research methodology tra...

Lecture 7 — Main Formula

The entire lecture can be compressed into:

Sample observations → Generalization → Uncertainty

and, where causal understanding is available:

Observation → Causal explanation → Stronger generalization

5. Lecture 8 — Deductive Logic Part 1

5.1 What Is Deduction?

Induction:

Particular → General

Deduction:

General → Particular

The lecturer emphasizes that a general premise must already be available.

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5.2 If–Then Structure

Many deductive arguments can be expressed as:

$A \rightarrow B$

Meaning:

If A occurs, B follows.

Example:

If copper is dipped in vinegar, it turns green.

Then:

$A = \text{copper dipped in vinegar}$

$B = \text{copper turns green}$

5.3 Modus Ponens

The first fundamental valid structure:

$$A \rightarrow B$$

$$A$$

Therefore:

$$\boxed{B}$$

This is **modus ponens**.  Week 2 research methodology tra...

Example

If:

All insects have six legs.

And:

This animal is an insect.

Then:

This animal has six legs.

5.4 A Very Common Logical Error

Consider:

$$A \rightarrow B$$

$$B$$

Therefore:

$$A$$

This is **invalid**.

It is the fallacy commonly called **affirming the consequent**.

Why?

Because B may have other causes.

The transcript's example is excellent:

If copper is dipped in vinegar $\rightarrow$ it turns green.

But:

Copper is green.

does not prove:

It was dipped in vinegar.

It might have been painted green.  Week 2 research methodology tra...

5.5 Another Common Error

Consider:

$$A \rightarrow B$$

$$\neg A$$

Therefore:

$$\neg B$$

This is also invalid.

It is the fallacy of **denying the antecedent**.

The transcript demonstrates:

Copper was not dipped in vinegar.

Therefore:

It cannot be green.

Wrong.

It could have become green another way.  Week 2 research methodology tra...

5.6 Modus Tollens

The second fundamental valid structure:

$$\begin{array}{c} A \rightarrow B \\ \neg B \end{array}$$

Therefore:

$\neg A$

This is **modus tollens**.  Week 2 research methodology tra...

Example

If:

If something is an insect $\rightarrow$ it has six legs.

And:

It does not have six legs.

Then:

It is not an insect.

Assuming the original premise is accepted without exceptions.

5.7 The Four Structures You Must Know

Structure	Valid?	Name
$A \rightarrow B, A \therefore B$		Modus Ponens
$A \rightarrow B, \neg B \therefore \neg A$		Modus Tollens
$A \rightarrow B, B \therefore A$		Affirming the consequent
$A \rightarrow B, \neg A \therefore \neg B$		Denying the antecedent

This is one of the most useful tables from the entire week.

5.8 Chains of Deduction

Real scientific reasoning rarely stops after one step.

You can have:

$$P_1 + P_2 \rightarrow C_1$$

then:

$$C_1 + P_3 \rightarrow C_2$$

then:

$$C_2 + P_4 \rightarrow C_3$$

and so on.

Eventually:

$$P_1, P_2, \dots, P_n \rightarrow C$$

The lecturer calls this a **chain of reasoning**.  Week 2 research methodology tra...

5.9 Why This Matters in Science

A scientific explanation often contains many intermediate mechanisms.

For example, the lecturer gives a long chain involving tree removal and climate effects:

Tree removal

↓

More light reaches ground

↓

Ground heats

↓

Air heats

↓

Air rises

↓

Atmospheric effects

↓

Rainfall changes

↓

Climate consequences



The point is not to memorize this particular causal chain.

The point is:

Complex conclusions can be constructed from many smaller logically connected steps.  Week 2 research methodology tra...

Lecture 8 — Mental Model

GENERAL PREMISE
↓
PARTICULAR FACT
↓
VALID LOGICAL RULE
↓
PARTICULAR CONCLUSION
↓
NEXT PREMISE
↓
NEXT CONCLUSION
↓
FINAL RESULT



6. Lecture 9 — Deductive Logic Part 2

This lecture introduces the complications that arise when the world is not simply:

0/1

or:

true/false.

6.1 Probabilistic Deduction

Instead of:

$$A \rightarrow B$$

we may have:

$$A \rightarrow B \text{ with probability } p$$

Example:

99.99% of humans have the heart on the left side of the chest.

Then:

$A = \text{person is human}$

gives:

$$P(B|A) = 0.9999$$

The lecturer calls this **probabilistic deductive logic**.  Week 2 research methodology tra...

6.2 Probabilistic Modus Ponens

Suppose:

$$P(B|A) = p$$

and:

$A = \text{true}$

Then the lecturer's reasoning is:

$$\boxed{P(B) = p}$$

or approximately the same level of confidence.

Example

If 95% of adult male sparrows fall within a weight range:

$$m_1 \leq W \leq m_2$$

then encountering a new adult male sparrow gives a prediction with approximately that confidence.

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6.3 Why Probabilistic Modus Tollens Is Different

Suppose:

$$P(B|A) = 0.95$$

You observe:

$$\neg B$$

Can you conclude:

$$\neg A?$$

No.

Why?

Because:

$$P(\neg B|A) = 0.05$$

is still possible.

So an adult male sparrow can legitimately fall outside the predicted range.

Therefore:

$$\boxed{\text{Probabilistic } A \rightarrow B + \neg B \not\Rightarrow \neg A}$$

The lecturer explicitly emphasizes this limitation.  Week 2 research methodology tra...

6.4 Important Modern Qualification

This part of the lecture should be understood carefully.

It is not that **probability makes modus tollens "invalid" in every possible formulation.**

Rather:

A probabilistic conditional does not provide the same certainty needed for classical modus tollens.

In probabilistic reasoning, you need to reason about:

$$P(A|B), \quad P(B|A), \quad P(A), \quad P(B)$$

and these cannot generally be inferred from one conditional probability alone.

This distinction will become very important when you later study statistics and Bayesian reasoning.

6.5 Defeasible Deductive Reasoning

Now we reach an interesting idea.

Sometimes:

$$A \rightarrow B$$

but then another premise says:

$$C \rightarrow \neg B$$

and both A and C are true.

Then:

$$B$$

and:

$$\neg B$$

appear to conflict.

What do we do?

The lecturer says:

Determine which premise is stronger.

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6.6 Medical Example

The lecturer uses a hypothetical case.

Rule 1

If heart blockage is above 95%, perform surgery.

Fact

Patient has >95% blockage.

Therefore:

Surgery.

But another rule:

Rule 2

If patient has haemophilia, surgery is unsafe.

Fact

Patient has haemophilia.

Therefore:

No surgery.

Now:

Surgery <i>vs</i> No surgery

The lecturer says the second premise is stronger in the hypothetical situation, so:

No surgery.

This is called **defeasible deductive reasoning** in the lecture.  Week 2 research methodology tra...

6.7 What "Defeasible" Means

The basic idea is:

A conclusion that is reasonable based on current information may have to be withdrawn when stronger/new information appears.

Therefore:

Conclusion $\xrightarrow{\text{new stronger premise}}$ Revision

This is very different from classical deduction, where a valid conclusion from fixed premises remains valid.

6.8 Scientific Ethics

The lecturer connects defeasible reasoning to scientific ethics.

Two competing premises might be:

Premise A

I should advance my career.

Premise B

Scientific knowledge should be used for human welfare and should not be used to cause serious harm.

The lecturer argues that B should override A when the two conflict.

He uses examples involving:

- food adulteration,
- biological weapons,
- technologies designed to harm humanity.

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Ethical principle

The lecturer's central position is:

Human welfare > career advancement

when the two genuinely conflict.

This is a methodological point, but also an ethical one.

6.9 A Deeper Research-Ethics Insight

This connects directly to what you will later encounter in:

- research integrity,
- responsible AI,
- dual-use research,
- safety,
- publication ethics,
- conflicts of interest.

A technically correct inference does not automatically tell you:

What ought I do with this knowledge?

That requires an additional ethical premise.

So:

Facts $\neq$ Values

Scientific reasoning tells you what follows from premises.

Ethics also asks:

Which premises ought we accept?

7. Lecture 10 — Syllogistic Logic Part 1

7.1 What Is a Syllogism?

Syllogistic logic is a formal system of deductive reasoning historically associated especially with Aristotle.

The lecturer's objective is to structure arguments so that valid and invalid logical forms can be distinguished.  Week 2 research methodology tra...

Basic structure:

Major premise + Minor premise $\rightarrow$ Conclusion

7.2 Major Premise

The general statement.

Example:

All mammals have hot blood.

7.3 Minor Premise

The particular classification.

A leopard is a mammal.

7.4 Conclusion

A leopard has hot blood.

So:

All mammals → hot blood

Leopard → mammal

Leopard → hot blood



The lecturer emphasizes that the general premise itself was obtained through induction.

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7.5 The Middle Term

This is extremely important.

Consider:

All mammals have hot blood.

Leopards are mammals.

Therefore, leopards have hot blood.

The term:

mammals

appears in both premises but disappears from the conclusion.

This is the:

Middle term

The conclusion connects:

subject of minor premise

to:

predicate of major premise.

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7.6 The Four Categorical Forms

The lecturer introduces four basic forms.

A — Universal affirmative

All A are B

E — Universal negative

No A are B

I — Particular affirmative

Some A are B

O — Particular negative

Some A are not B

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7.7 Why A/E/I/O?

They are conventional abbreviations from the traditional categorical-logic system.

The lecturer represents them symbolically so that the actual content can be substituted later.

For example:

$$A(a, b)$$

can represent:

All A are B.

Then the specific categories can be substituted.

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7.8 Example — Physics

Major

All spin-half particles are fermions.

Minor

All electrons are spin-half particles.

Conclusion

All electrons are fermions.

The middle term:

spin-half particles

disappears from the conclusion.

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7.9 Chains of Syllogisms

You can chain them:

Lions → big cats

big cats → predators

predators → carnivores

Therefore:

Lions → carnivores

The lecturer explicitly gives this example. 📄 Week 2 research methodology tra...

8. Lecture 11 — Syllogistic Logic Part 2

This lecture becomes much more practical.

8.1 Venn Diagrams as a Logic Tool

Instead of memorizing all valid syllogisms, the lecturer proposes using **Venn diagrams**.

The basic idea:

Represent each category as a region.

Then:

- remove impossible regions,
- mark regions that must contain something,
- inspect what remains logically possible.

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8.2 Representing "All A Are B"

All A are B

means:

$$A - B = \emptyset$$

The portion of A outside B must contain nothing.

The lecturer represents this by blocking/hatching the region.  Week 2 research methodology tra...

8.3 Representing "No A Is B"

No A is B

means:

$$A \cap B = \emptyset$$

The intersection must be empty.

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8.4 Representing "Some A Are B"

Some A are B

means:

$$A \cap B \neq \emptyset$$

The intersection must contain at least one element.

The lecturer uses an **X/cross** to indicate existence.  Week 2 research methodology tra...

8.5 Representing "Some A Are Not B"

Some A are not B

means:

$$A - B \neq \emptyset$$

Again, the lecturer places an X in the appropriate region.  Week 2 research methodology tra...

8.6 The Critical Rule

This is one of the best practical lessons of Lecture 11:

Do not infer information that the premises do not provide.

Suppose:

Some A are B

and:

Some B are C

Can we conclude:

Some A are C?

No.

Why?

Because the $A \cap B$ individuals need not be the same as the $B \cap C$ individuals.

Therefore:

$$A \cap B \neq \emptyset \wedge B \cap C \neq \emptyset \not\Rightarrow A \cap C \neq \emptyset$$

The lecturer explicitly warns against making this assumption.  Week 2 research methodology tra...

8.7 Certainty vs Possibility

This distinction is extremely important.

Suppose the premises allow:

A and C overlap.

That does **not** mean:

A and C definitely overlap.

So:

Guaranteed conclusion

Must be true in **every** diagram consistent with the premises.

Possible conclusion

True in **at least one** diagram consistent with the premises.

This is a powerful way to think about logical inference.

8.8 Venn Diagram Procedure

For categorical syllogisms:

Step 1

Identify the three categories.

Step 2

Draw the three overlapping sets.

Step 3

Translate each premise into constraints.

Step 4

Block impossible regions.

Step 5

Place X where existence is required.

Step 6

Ask:

Is the proposed conclusion true in **every possible configuration** consistent with the premises?

If yes:

Valid

If no:

Invalid

This is much more useful than memorizing dozens of syllogistic forms.

8.9 Example

Premise 1

All men are mortal.

Premise 2

All Greeks are men.

Therefore:

All Greeks are mortal.

The Venn diagram forces:

$$Greek \subseteq Men \subseteq Mortal$$

Therefore:

$$Greek \subseteq Mortal$$

The conclusion is guaranteed.

The lecturer walks through this construction explicitly.  Week 2 research methodology tra...

8.10 Another Example

Premise

No reptiles have fur.

Premise

All snakes are reptiles.

Therefore:

No snakes have fur.

Because:

$$Snake \subseteq Reptile$$

and:

$$Reptile \cap Fur = \emptyset$$

therefore:

$$Snake \cap Fur = \emptyset$$

Valid.  Week 2 research methodology tra...

8.11 Invalid Inference Example

Premise

All tigers are hunters.

Premise

Some carnivores are hunters.

Proposed conclusion:

Some carnivores are not tigers.

The lecturer demonstrates that this is **not guaranteed**.

Why?

The carnivorous hunters could potentially all be tigers.

The premises don't force the existence of a carnivore outside the tiger set.

Therefore:

Invalid inference

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9. Lecture 12 — Truth vs Validity

This is the philosophical culmination of the week.

9.1 Truth

Truth concerns whether a proposition corresponds to the relevant facts/reality.

Example:

"Electrons have spin $1/2$."

This can be scientifically evaluated as true/false.

9.2 Validity

Validity concerns the **structure of an argument**.

An argument is valid when:

If the premises were true, the conclusion would necessarily follow.

Validity does **not** ask whether the premises actually are true.

9.3 The Classic Distinction

Consider:

All dinosaurs are mammals.

False.

All cats are dinosaurs.

False.

Therefore:

All cats are mammals.

The conclusion happens to be true.

But the argument can still have a valid logical structure.

The lecturer uses essentially this example to demonstrate that false premises can lead through valid logic to a true conclusion.  Week 2 research methodology tra...

9.4 The Four Possibilities

Premises	Logic	Conclusion
True	Valid	Must be true
False	Valid	Could be true or false
True	Invalid	Could be true or false
False	Invalid	Could be true or false

The critical case is:

True conclusion does not prove valid reasoning

9.5 Validity Is About Necessity

The best mental definition:

Could the premises all be true while the conclusion is false?

If yes:

Invalid

If no:

Valid

This is exactly what the Venn-diagram method is testing.

9.6 Truth Does Not Validate Logic

The lecturer gives a particularly useful example.

Premise 1

All particles with integer-valued spin are bosons.

Premise 2

Electrons do not have integer-valued spin.

Conclusion

Electrons are not bosons.

The statements are scientifically true.

But the logical structure is not necessarily valid merely because the conclusion happens to be true.

The lecturer constructs a Venn diagram to show that the premises don't logically exclude all possibilities required for the conclusion.  Week 2 research methodology tra...

This is an excellent lesson:

Do not use your knowledge of reality to decide whether an argument is logically valid.

Evaluate the structure independently.

10. Scientific Hypothesis Testing and Logic

Now Week 2 connects directly to actual research.

Suppose:

Hypothesis

$$H$$

From H we derive:

$$H \rightarrow P$$

where P is an observable prediction.

Then:

Experiment

Observe:

$$\neg P$$

Therefore, under the assumed logical structure:

$$\boxed{\neg H}$$

This is the logic of **modus tollens**.

The lecturer explicitly explains this in the context of scientific hypotheses and postulates.

 Week 2 research methodology tra...

10.1 Why Confirmation Is Weaker

Suppose:

$$H \rightarrow P$$

and experiment gives:

$$P$$

Can you conclude:

$$H?$$

No.

Because:

$$P$$

may have other explanations.

This is the same error as:

$$A \rightarrow B, \quad B \therefore A$$

which is affirming the consequent.

The lecturer explicitly connects this point to scientific testing.

 Week 2 research methodology tra...

10.2 Falsification Logic

This gives a useful asymmetry:

Failed prediction

$$H \rightarrow P$$

$$\neg P$$

$$\therefore \neg H$$

This can be logically powerful.

Successful prediction

$$H \rightarrow P$$

$$P$$

$$\therefore H$$

This is **not** logically valid.

Therefore:

Failure of prediction can strongly challenge a hypothesis; success does not un

Important qualification

In real science, even $\neg P$ does not always immediately establish $\neg H$, because the prediction may depend on auxiliary assumptions:

$$H + A_1 + A_2 + A_3 \rightarrow P$$

If P fails, the problem could lie in:

- H,
- A_1 ,
- A_2 ,
- A_3 ,
- measurement,
- experimental implementation.

This is the **Duhem–Quine problem** / underdetermination issue.

The lecture simplifies this, but the underlying logical lesson remains valuable.

11. Why Induction Comes Before Deduction in Scientific Practice

The lecturer argues:

Deduction needs a general premise.

Where does the general premise come from?

Usually:

Induction

Therefore:

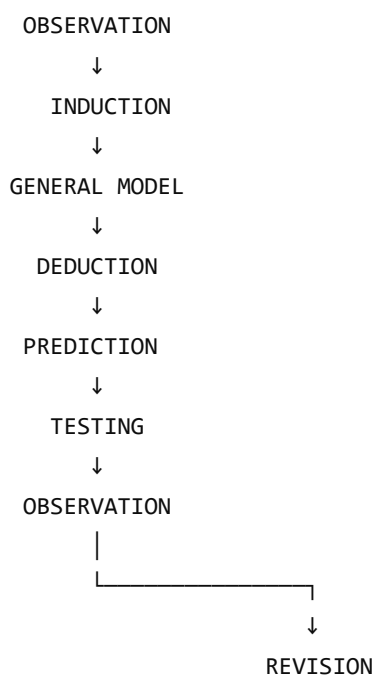
Induction → General premise → Deduction

The lecturer therefore calls induction **prior** to deduction in the scientific workflow.

📄 Week 2 research methodology tra...

11.1 But They Form a Loop

This is even more important than the linear sequence.



Science therefore isn't:

Induction OR deduction.

It is:

Induction + Deduction + Testing

The lecturer explicitly describes science as thriving on their joint application.

📖 Week 2 research methodology tra...

12. Astronomy Example — Cepheid Variables

This is arguably the best example in Week 2 because it demonstrates the full induction → deduction cycle.

12.1 The Problem

How do we measure distances to very distant stars and galaxies?

Nearby stars can be measured using **parallax**.

The basic geometric idea:

$$d = \frac{B}{\tan \theta}$$

where:

- B = baseline
- θ = parallax angle

The lecturer describes using Earth's orbital position as the effective baseline for stellar parallax. 📖 Week 2 research methodology tra...

But eventually stars are too distant for useful parallax measurements.

Science reaches a measurement barrier.

12.2 Henrietta Leavitt

Henrietta Leavitt studied **Cepheid variable stars**.

These stars vary periodically in brightness.

The lecturer explains that she studied Cepheids whose distances could be independently measured and investigated the relationship between:

period

and:

absolute luminosity.

📄 Week 2 research methodology tra...

12.3 The Inductive Step

From the observed Cepheids:

$$P_1 \rightarrow L_1$$

$$P_2 \rightarrow L_2$$

$$P_3 \rightarrow L_3$$

etc.

A general relationship was inferred:

$$L = f(P)$$

The lecturer explicitly describes this as an inductive generalization.

📄 Week 2 research methodology tra...

12.4 The Deductive Step

Now observe a distant Cepheid.

Measure:

$$P$$

Use the inferred relationship:

$$P \rightarrow L$$

Then measure apparent luminosity:

$$L_{\text{apparent}}$$

From the relationship between apparent and absolute luminosity, infer distance.

Conceptually:

$$P \rightarrow L_{\text{absolute}} \rightarrow d$$

This is deduction applied to a particular star.

12.5 Hubble and Andromeda

The lecturer then explains how the method allowed Edwin Hubble to identify Cepheids in what had previously been regarded as the Andromeda "nebula" and estimate its distance.

The important methodological point is:

Induction creates a general relationship

then:

Deduction applies it to new individual cases

 Week 2 research methodology tra...

12.6 The Inductive Error

Later, the lecturer notes that Cepheids were not simply one homogeneous class.

Different types of Cepheids have different relationships.

Therefore the original generalization had to be refined.

This illustrates:

Inductive conclusion → New evidence → Correction

The lecturer explicitly uses this as an example of why scientists must retain doubt and recheck conclusions.  Week 2 research methodology tra...

13. The Most Important Lesson From the Cepheid Example

Scientific knowledge often follows:

Limited observations → General relationship → Application to new cases →



That is **science in action**.

It is not:

"Scientists discover the perfect law once and forever."

It is:

Build the best general model available, test it against new reality, and revise it when reality exposes its limitations.

14. Fact vs Interpretation vs Inference

[SOURCE — DIRECTLY FROM LECTURE]

Inductive reasoning

Particular → general.  Week 2 research methodology tra...

Deductive reasoning

General → particular.  Week 2 research methodology tra...

Inductive leap

Generalization exceeds the finite observations actually made.  Week 2 research methodology tra...

Modus ponens

$$A \rightarrow B, \quad A \therefore B$$

 Week 2 research methodology tra...

Modus tollens

$$A \rightarrow B, \quad \neg B \therefore \neg A$$

 Week 2 research methodology tra...

Probabilistic deduction

General probabilistic premise → probabilistic conclusion.  Week 2 research methodology tra...

Syllogism

Major premise + minor premise → conclusion.  Week 2 research methodology tra...

Truth ≠ validity

The truth of a conclusion does not establish the validity of the reasoning.

 Week 2 research methodology tra...

15. Important Corrections / Qualifications

This section is especially important because you asked for the same kind of deep treatment as Week 1.

15.1 Empirical induction $\neq$ mathematical induction

The lecturer groups mathematical induction with complete enumeration.

But mathematically:

$$P(1)$$

and:

$$P(n) \rightarrow P(n + 1)$$

constitute a **deductive proof**.

It does not depend on observing infinitely many cases.

So don't internalize:

"Mathematical induction is empirical induction."

It isn't.

15.2 "Proper induction = causal connection" is too strong

A causal mechanism can strengthen an inductive inference enormously.

But not every scientifically legitimate inductive inference requires a fully understood causal mechanism.

For example, statistical estimation can be scientifically useful even when the underlying causal mechanism is not fully understood.

So:

Causal understanding can strengthen induction, but it is not a universal prerequisite for scientific induction.

15.3 Probabilistic reasoning is richer than "confidence"

The lecturer sometimes speaks of probability as directly representing "confidence."

That is useful pedagogically, but technically:

$$P(\text{data}|\text{hypothesis})$$

is not necessarily the same as:

$$P(\text{hypothesis}|\text{data})$$

This distinction becomes crucial in Bayesian inference.

So later, when you study statistics, be careful not to equate every probability statement with personal confidence.

15.4 Scientific conclusions are not simply "true/false"

Scientific propositions can have:

- uncertainty,
- measurement error,
- model dependence,
- statistical uncertainty,
- limited domains of applicability.

Therefore real research often asks:

How strongly does the evidence support this claim?

rather than only:

True or false?

15.5 A failed prediction does not automatically destroy a hypothesis

The lecturer's simplified logical structure is:

$$H \rightarrow P$$

$$\neg P$$

therefore:

$$\neg H$$

This is logically valid if **H really entails P** and the premises are correctly specified.

In actual research:

$$H + A_1 + A_2 + A_3 \rightarrow P$$

may be the real structure.

If P fails, you need to determine whether the problem lies in:

- H,
- auxiliary assumptions,
- measurement,
- experimental design,
- implementation.

This is why good research investigates failure rather than immediately declaring a theory dead.

15.6 "Valid logic" does not mean "scientifically useful"

An argument can be logically valid while starting from ridiculous premises.

For example:

All dragons are invisible.
Socrates is a dragon.
Therefore Socrates is invisible.

The structure could be valid.

The premises are not established facts.

Therefore:

Logical validity $\neq$ empirical truth

This distinction is one of the major purposes of Lecture 12.

16. AI/ML Research Connections

This week is **extremely relevant** to your goal of becoming an applied AI researcher.

16.1 Induction in Machine Learning

Machine learning is fundamentally an inductive activity.

You observe:

$$D = \{(x_i, y_i)\}_{i=1}^n$$

and try to infer a general function:

$$f : X \rightarrow Y$$

Then you apply f to unseen data.

So:

Training examples $\rightarrow$ learned generalization

is essentially an inductive leap.

16.2 Generalization = Inductive Leap

Suppose your model sees:

10,000

training images.

It learns a classifier.

Then you evaluate on:

new images

You are assuming that the patterns learned from the sample generalize to the broader data-generating distribution.

That is:

Sample $\rightarrow$ Population/distribution

This is precisely the sample-to-population problem discussed in Lecture 7.

16.3 Overfitting Is an Inductive Failure

Suppose:

$$\textit{Training Accuracy} = 99.9\%$$

but:

$$\textit{Test Accuracy} = 72\%$$

The model failed to generalize.

In logical terms:

It learned a pattern that worked for the observed cases but did not reliably extend to new cases.

That is exactly the danger of naive induction.

16.4 Training Data $\neq$ Reality

Suppose your training dataset contains:

mostly ripe red apples.

The model may learn:

$$\textit{red} \rightarrow \textit{ripe}$$

instead of:

$$\textit{actual visual maturity cues} \rightarrow \textit{ripe}$$

Then:

green but ripe apples

cause failure.

This is analogous to the classic:

"I've seen only white swans."

problem.

16.5 Dataset Bias

If:

$$P_{\text{train}}(X, Y) \neq P_{\text{deployment}}(X, Y)$$

then the inductive generalization may fail.

This is why AI researchers need to think about:

- distribution shift,
 - domain shift,
 - covariate shift,
 - label shift,
 - dataset representativeness.
-

16.6 Deduction in AI Research

Suppose your learned/general theoretical premise is:

Increasing model capacity under these conditions reduces approximation error.

Then for a particular architecture:

A

you derive a prediction:

P

and experimentally test it.

That is:

General principle $\rightarrow$ specific prediction

16.7 Modus Ponens in ML

Suppose:

If model uses method $X \rightarrow$ performance improves under condition C

Then:

Our model uses X under C .

Therefore:

We expect improved performance.

This is a deductive prediction.

16.8 Affirming the Consequent in ML

Suppose:

$$X \rightarrow \text{higher accuracy}$$

You observe:

higher accuracy

Then concluding:

Therefore, X caused the improvement.

is invalid.

Maybe:

- data augmentation caused it,
- larger compute caused it,
- hyperparameter tuning caused it,
- leakage caused it,
- random variation caused it.

This is **exactly** the logical mistake from Lecture 8.

16.9 Ablation Studies

Suppose:

$$Model = A + B + C$$

and:

$$Performance = 95\%$$

Remove B:

$$Model = A + C$$

Performance becomes:

90%

This gives evidence that B contributes to performance.

But even here, you should ask:

Is B independently responsible, or does B interact with A/C?

Again:

logical reasoning must be combined with experimental design.

16.10 Syllogism in Research Papers

A surprising amount of research writing has this structure.

General premise

Existing methods perform poorly under low-compute conditions.

Specific premise

Our deployment environment is low-compute.

Conclusion

Therefore, conventional methods may be unsuitable for our deployment environment.

But then the **premises themselves require evidence**.

That is the crucial distinction.

17. Researcher Mistakes Week 2 Helps Prevent

Mistake	Logical problem
"It worked on 100 examples, so it always works."	Overgeneralization
"The benchmark improved, therefore our explanation is correct."	Affirming the consequent
"We didn't observe X, therefore the hypothesis is false."	May ignore auxiliary assumptions

Mistake	Logical problem
"The conclusion is true, therefore the argument is valid."	Truth $\neq$ validity
"A famous paper says X, therefore X is true."	Authority substituted for evidence
"A and B overlap, therefore A causes B."	Correlation/causality confusion
"The model performs well on training data, therefore it generalizes."	Inductive failure/overfitting
"Two groups share B, therefore they overlap."	Invalid syllogistic inference
"A possible conclusion must be true."	Possibility $\neq$ necessity
"Probability = certainty."	Misunderstanding uncertainty
"One experiment disproves everything."	Ignores auxiliary assumptions
"One successful experiment proves the theory."	Confirmation $\neq$ deductive proof

18. Core Concept Connections

Induction and Deduction

Induction : Particular $\rightarrow$ General

Deduction : General $\rightarrow$ Particular

Scientific cycle

Observation $\rightarrow$ Induction $\rightarrow$ Hypothesis $\rightarrow$ Deduction $\rightarrow$ Prediction $\rightarrow E$



Classical logic

$$A \rightarrow B$$

Modus Ponens

$$A \Rightarrow B$$

Modus Tollens

$$\neg B \Rightarrow \neg A$$

Invalid structures

$$B \Rightarrow A$$

✗ Affirming the consequent.

$$\neg A \Rightarrow \neg B$$

✗ Denying the antecedent.

19. Syllogism Cheat Sheet

A

All A are B

$$A \subseteq B$$

E

No A are B

$$A \cap B = \emptyset$$

I

Some A are B

$$A \cap B \neq \emptyset$$

O

Some A are not B

$$A - B \neq \emptyset$$

20. Venn Diagram Logic

"All A are B"

Block:

$$A \setminus B$$

"No A are B"

Block:

$$A \cap B$$

"Some A are B"

Put X in:

$$A \cap B$$

"Some A are not B"

Put X in:

$$A \setminus B$$

21. The Most Important Venn Diagram Rule

Suppose premises permit two possibilities:

World 1

$$A \cap C \neq \emptyset$$

World 2

$$A \cap C = \emptyset$$

Then you **cannot conclude**:

$$A \cap C \neq \emptyset$$

because the conclusion isn't true in every permitted world.

Therefore:

Validity requires necessity, not mere possibility.

This is probably the best way to understand the Venn-diagram exercises.

22. Truth vs Validity — Ultimate Mental Model

Think of two separate questions.

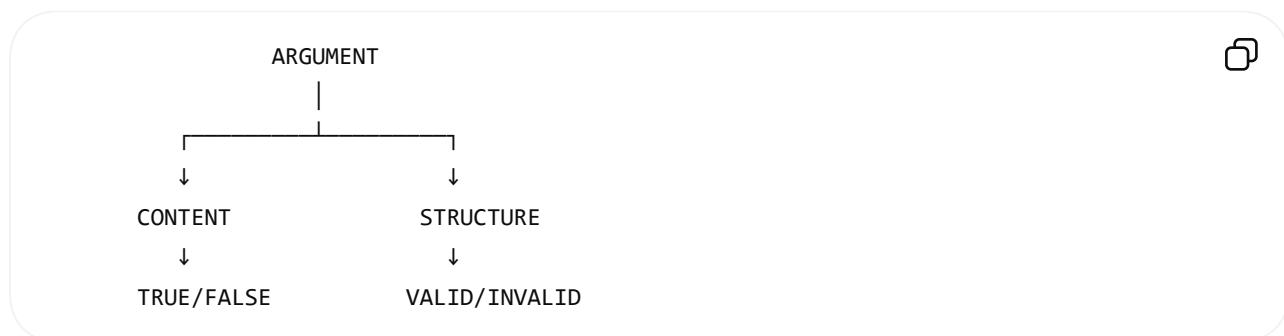
Question 1 — Truth

Is the statement actually true?

Question 2 — Validity

Does the conclusion necessarily follow from the premises?

You can therefore have:



Never mix the two.

23. Scientific Hypothesis Logic

A useful research abstraction is:

$$H + A \rightarrow P$$

where:

- H = hypothesis
- A = auxiliary assumptions
- P = prediction

Then:

Prediction fails

$$\neg P$$

→ something in $H + A$ must be reconsidered.

Prediction succeeds

$$P$$

→ $H + A$ survives the test, but is not uniquely proven.

This is much closer to how real scientific research works than the simplistic:

"Experiment proved my hypothesis."

24. Knowledge Gaps

Topic	Priority
Basic induction/deduction distinction	● Essential
Modus ponens/tollens	● Essential
Affirming consequent / denying antecedent	● Essential
Truth vs validity	● Essential
Syllogistic A/E/I/O	● Important
Venn-diagram validity testing	● Important
Sample vs population	● Essential for research
Probability and confidence	● Essential
Problem of induction	● Important philosophical foundation
Bayesian reasoning	● Later essential
Modern causal inference	● Later essential for AI research
Formal logic beyond syllogisms	● Useful later

25. Critical Thinking Questions

1. Why can't finite observations prove a universal scientific statement?

Because there can always be an unobserved case that violates the generalization.

2. Why is induction necessary if it is uncertain?

Because science must generalize beyond observed cases to make predictions and construct theories.

3. Why isn't deduction enough?

Because deduction requires premises.

Science needs a way of obtaining useful general premises from empirical investigation.

4. Why isn't induction enough?

Because once we have a general hypothesis, we need deduction to derive specific predictions that can be tested.

5. Why is modus ponens valid?

Because if:

$$A \rightarrow B$$

and:

$$A$$

then B necessarily follows according to the conditional premise.

6. Why is affirming the consequent invalid?

Because:

$$A \rightarrow B$$

doesn't mean:

$$B \rightarrow A$$

B can have multiple explanations.

7. Why is truth different from validity?

Truth describes propositions.

Validity describes the relationship between premises and conclusion.

8. Why can a false premise produce a true conclusion?

Because the conclusion may happen to be true independently of the premises.

9. Why are Venn diagrams useful?

Because they let you test whether the conclusion is true across **all configurations compatible with the premises**.

10. Why is the Cepheid example important?

Because it demonstrates the actual scientific cycle:

Induction → general relationship → deduction → new application → revision



26. Beyond the Lecture — A Deeper Research Perspective

There is a subtle but extremely important idea hiding underneath the whole week:

Science is not simply about collecting more observations. It is about constructing generalizations that survive attempts to break them.

Imagine two researchers.

Researcher A

Collects:

10,000

examples supporting their hypothesis.

Researcher B

Collects:

100

examples, but deliberately searches for:

- edge cases,
- counterexamples,
- distribution shifts,
- competing explanations,
- failure conditions.

Researcher B may actually be doing **better science**.

Why?

Because scientific knowledge grows not merely by confirmation but by discovering where our generalizations stop working.

This is exactly what happened in the Cepheid example discussed in Lecture 12.

 Week 2 research methodology tra...

27. Connection to Your Research Career

This week gives you a very useful framework for evaluating your own research projects.

Suppose you claim:

"My proposed model improves fine-grained classification."

Don't stop there.

Ask:

Inductive question

What observations justify the general claim?

Deductive question

What predictions should follow if the claim is correct?

Validity question

Does the conclusion actually follow from the evidence?

Alternative-explanation question

Could another mechanism explain the result?

Generalization question

Does it work outside the training distribution?

Falsification question

What result would make me reconsider the claim?

Reproducibility question

Does the result survive independent repetition?

That is the mindset of a researcher rather than simply a model builder.

28. AI Research Example — Full Logic

Suppose you hypothesize:

$H =$ Progressive transfer learning improves generalization

You derive:

$$H \rightarrow P$$

where:

$P =$ Progressive transfer learning beats conventional transfer learning on uns



You conduct the experiment.

Case A

$$P = \text{true}$$

You have evidence supporting H.

But:

$$P \not\Rightarrow H$$

because perhaps another factor caused the improvement.

Case B

$$P = \text{false}$$

Now H is challenged.

But investigate:

- same dataset?
- same preprocessing?
- same training budget?
- same hyperparameters?
- same random seeds?
- implementation correct?
- evaluation correct?

Only after examining those auxiliary assumptions should you determine what exactly failed.

29. Common Traps

Trap 1

"I observed it repeatedly, so it is proven."

No. That's induction.

Trap 2

"The conclusion is true, so the argument must be valid."

No.

Truth $\neq$ validity.

Trap 3

"B happened, therefore A caused it."

No.

You may be affirming the consequent.

Trap 4

"A didn't happen, therefore B didn't happen."

No.

You may be denying the antecedent.

Trap 5

"There could be an $A \cap C$ overlap, therefore there is an $A \cap C$ overlap."

No.

Possibility $\neq$ necessity.

Trap 6

"95% confidence means there's a 95% chance my particular hypothesis is true."

Not necessarily. The precise interpretation depends on what probability statement is being made.

Trap 7

"One counterexample destroys every aspect of a theory."

Not necessarily.

It may invalidate:

- a universal formulation,
- a particular auxiliary assumption,
- an application,

- or the measurement.

You must identify **what exactly failed**.

30. One-Page Revision Sheet

INDUCTION

Particular $\rightarrow$ General

- Uses observations.
 - Contains an inductive leap.
 - Generalizes beyond observed cases.
 - Therefore involves uncertainty.
 - Can be deterministic or probabilistic.
 - Can concern qualitative or quantitative properties.
-

DEDUCTION

General $\rightarrow$ Particular

Requires a premise.

MODUS PONENS

$$A \rightarrow B$$

$$A$$

$$\therefore B$$

 Valid.

MODUS TOLLENS

$$A \rightarrow B$$

$$\neg B$$

$$\therefore \neg A$$

✓ Valid for classical conditionals.

AFFIRMING CONSEQUENT

$$A \rightarrow B$$

$$B$$

$$\therefore A$$

✗ Invalid.

DENYING ANTECEDENT

$$A \rightarrow B$$

$$\neg A$$

$$\therefore \neg B$$

✗ Invalid.

SYLLOGISM

$Major\ premise + Minor\ premise \rightarrow Conclusion$
--

Middle term disappears.

A/E/I/O

Symbol	Meaning
A	All A are B
E	No A are B
I	Some A are B
O	Some A are not B

VALIDITY

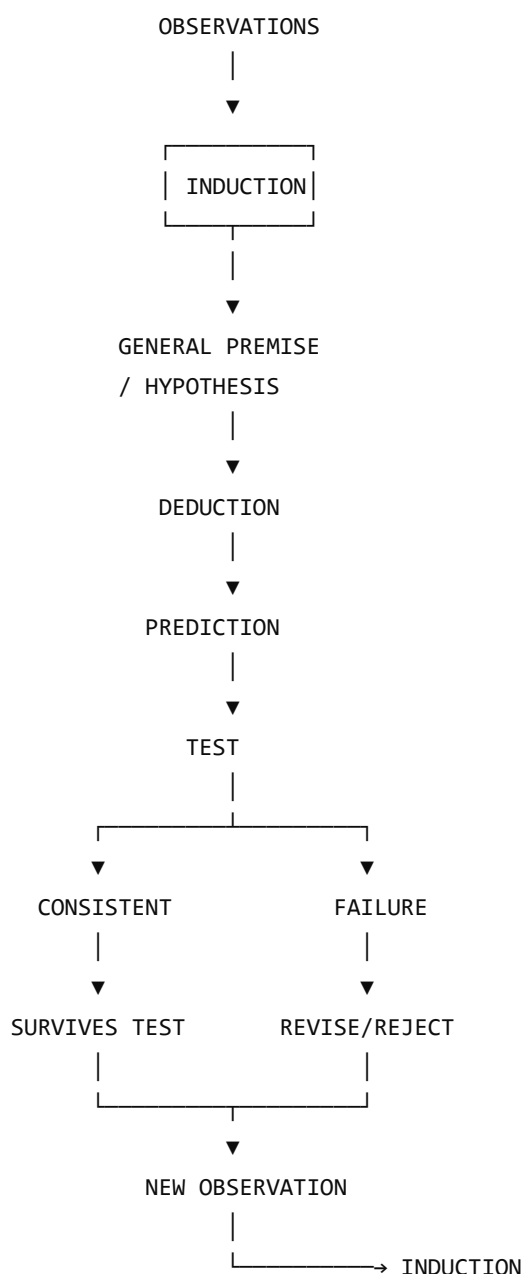
Does the conclusion necessarily follow from the premises?

TRUTH

Does the proposition correspond to reality?

31. Final Mental Model

If you remember only one diagram from Week 2, make it this:



And underneath the entire process:

VALIDITY $\neq$ TRUTH

and:

CORRELATION $\neq$ CAUSATION

and:

CONFIRMATION $\neq$ PROOF

32. What Week 2 Really Adds to Week 1

Week 1 taught you:

What scientific inquiry is and why objective, skeptical investigation matters.

Week 2 teaches you:

How reasoning actually moves through that investigation.

So the combined framework becomes:

Reality $\rightarrow$ Observation $\rightarrow$ Question $\rightarrow$ Induction $\rightarrow$ Hypothesis $\rightarrow$ Deduction

That is a **very important foundation for research methodology.**

And for you specifically as someone moving toward AI research, the most valuable transformation is:

Don't merely ask whether your experiment produced a good result. Ask what exact proposition the result supports, what assumptions connect the result to that proposition, what alternative explanations exist, and what experiment would distinguish them.

That is the point where experimentation starts becoming **research** rather than simply model development.

Sources

      Convert to PDF